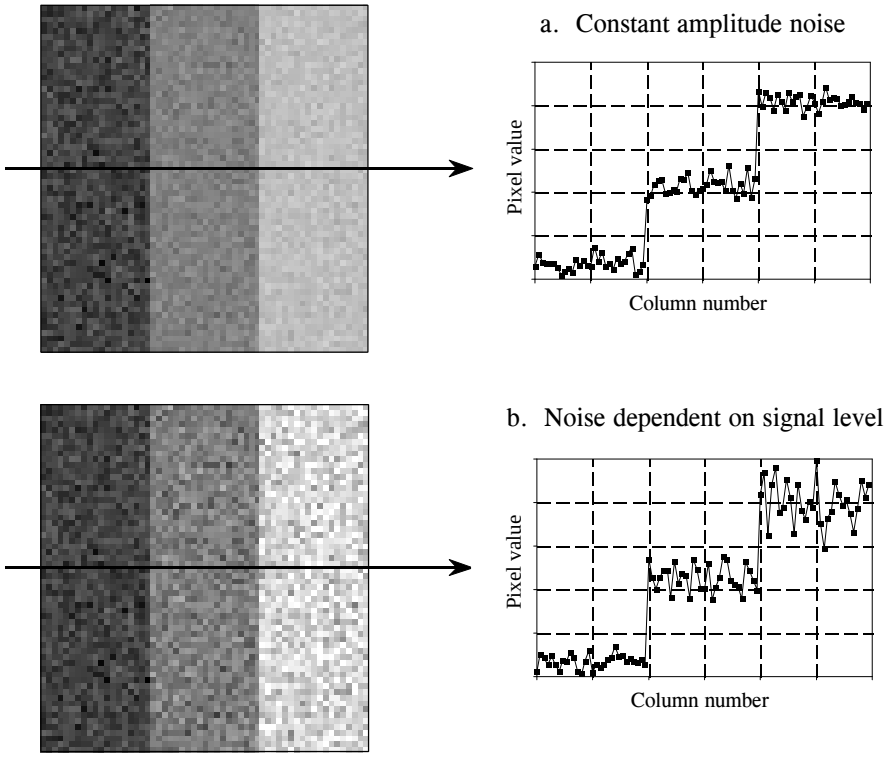

FIGURE 25-9
Image noise. Random noise in images takes two general forms. In (a), the amplitude of the noise remains constant as the signal level changes. This is typical of electronic noise. In (b), the amplitude of the noise increases as the square-root of the signal level. This type of noise originates from the detection of a small number of particles, such as light photons, electrons, or x-rays.

well-to-well. A key feature of Poisson statistics is that the standard deviation is equal to the square-root of the number of individual particles. That is, if there are N particles in each pixel, the mean is equal to N and the standard deviation is equal to $\sqrt{N}$. This makes the signal-to-noise ratio equal to $N/\sqrt{N}$, or simply, $\sqrt{N}$. In equation form:

EQUATION 25-1
Poisson statistics. In a Poisson distributed signal, the mean, μ, is the average number of individual particles, N. The standard deviation, σ, is equal to the square-root of the average number of individual particles. The signal-to-noise ratio (SNR) is the mean divided by the standard deviation.

$$\mu = N$$

$$\sigma = \sqrt{N}$$

$$SNR = \sqrt{N}$$

In the CCD example, the standard deviation is $\sqrt{10{,}000} = 100$. Likewise the signal-to-noise ratio is also $\sqrt{10{,}000} = 100$. If the average number of electrons per well is increased to one million, both the standard deviation and the SNR increase to 1,000. That is, the noise becomes larger as the signal becomes